SEMINAR 10

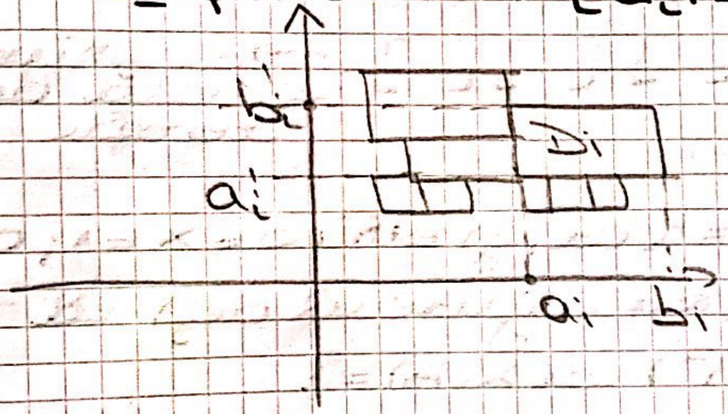
Mulțimi măsurabile Jordan

• **Caz particular $\mathbb{R}^2$.**

Def: Numim mulțime elementară în $\mathbb{R}^2$, orice mulțime $E \subseteq \mathbb{R}^2$ care poate fi scrisă ca reuniune finită de dreptunghiuri cu laturile paralele cu axele de coordonate, cu interioare disjuncte două câte două, ie

$$E = \bigcup_{i \in I} D_i, \text{ cu } \mathring{D}_j \cap \mathring{D}_k = \emptyset, \forall j \neq k$$

I finită $\leadsto [a_i, b_i] \times [a_i', b_i']$



$$vol(D_i) = Aria(D_i) = (b_i - a_i)(b_i' - a_i')$$

• Definim aria mulț E (sau volumul lui E):

$$vol(E) := v(E) = \sum_{i \in I} v(D_i)$$

Def: Pt. o mulțime $M \subseteq \mathbb{R}^2$ definim:

$\lambda_*(M) = \sup_{E \subset M, E \in \mathcal{Q}} v(E)$ - măsura (aria) interioară

$\lambda^*(M) = \inf_{M \subset E, E \in \mathcal{Q}} v(E)$ - măsura exterioară

Def. Spunem că $M \subseteq \mathbb{R}^2$ este măsurabilă Jordan (sau are arie), dacă cele 2 măsuri sunt egale.

În acest caz:

$$\lambda(M) := \lambda_*(M) = \lambda^*(M)$$

↳ măsura Jordan

Scriem $M \in \mathcal{J}(\mathbb{R}^2)$.

Obs. • $E \subseteq \mathbb{R}^2$ elementară $\Rightarrow E \in \mathcal{J}(\mathbb{R}^2)$ și $\lambda(E) = vol(E)$

• $\emptyset \in \mathcal{J}(\mathbb{R}^2)$ și $\lambda(\emptyset) = 0$.

Analog se va defini măs. Jordan pe $\mathbb{R}$, $\mathbb{R}^n$

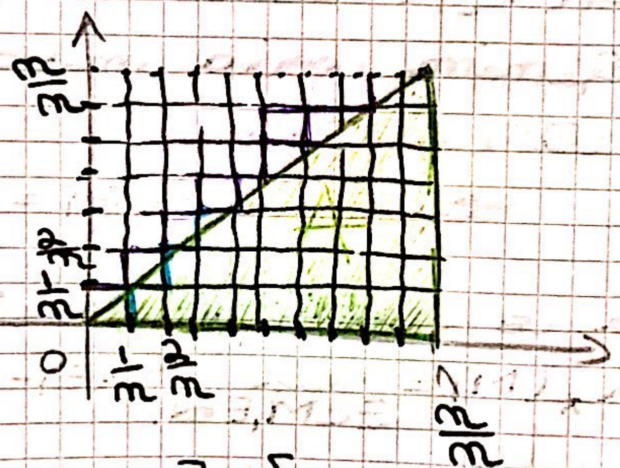
↓ Elementare din $\mathbb{R}$:

$[\ ]\ [\ ]\ [\ ]$ → reuniune finită de intervale

$\boxed{\text{Ex1}}$ Fie $A = \{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 1;\ 0 \le y \le x\}$

Dem (cu ajutorul def) că $A \in \mathcal{J}(\mathbb{R}^2)$ cu $\lambda(A) = ?$

Sol:



$$D_{ij} = [x_{i-1}, x_i] \times [y_{i-1}, y_i]$$

$$\left[\frac{i-1}{n}, \frac{i}{n}\right] \times \left[\frac{j-1}{n}, \frac{j}{n}\right]. \quad 1 \le i \le j \le n$$

Fie $\Delta_n = \{D_{ij} \mid 1 \le i, j \le n\}$ diviziune a lui $I = [0,1] \times [0,1]$

Fie $E_n = \bigcup_{1 \le j < i \le n} D_{ij} \in \mathcal{E}(\mathbb{R}^2)$ și $E_n \subset A$

Fie $F_n = \bigcup_{1 \le j \le i \le n} D_{ij} \in \mathcal{E}(\mathbb{R}^2)$ și $F_n \supset A$.

Avem $\lambda(E_n) = v(E_n) = \sum_{1 \le j < i \le n} D_{ij}$

$= (n-1 + n-2 + \dots + 2 + 1) \cdot \frac{1}{n^2} =$

$= \frac{(n-1) \cdot n}{2} \cdot \frac{1}{n^2} = \frac{n-1}{2n} = \frac{1}{2}\left(1 - \frac{1}{n}\right)$.

- $\lambda(F_n) = v F_n = \sum_{1 \le j \le i \le n} D_{ij} =$

$= (n + (n-1) + \dots + 2 + 1) \cdot \frac{1}{n^2} =$

$= \frac{n(n+1)}{2} \cdot \frac{1}{n^2} = \frac{1}{2}\left(1 + \frac{1}{n}\right)$.

Deci

$$\underbrace{\sup_n \underbrace{v(E_n)}_{\lambda(E_n)}}_{=\frac{1}{2}} \le \sup_{\substack{E \subset A \\ E\, el.}} v(E) = \lambda_*(A) \le$$

$$\le \lambda^*(A) = \inf_{\substack{F \supset A \\ F\, elem.}} v(F) \le \underbrace{\inf_n \underbrace{v(F_n)}_{\lambda(F_n)}}_{=\frac{1}{2}}.$$

$\Downarrow$

$$A \in \mathcal{J}(\mathbb{R}^2),\ \lambda(A) = \frac{1}{2}.$$

T (curs)

Fie $A \subseteq \mathbb{R}^n$ mărginită. Atunci

UASE:

1) $A \in \mathcal{J}(\mathbb{R}^n)$

2) $\mathring{A}, \overline{A} \in \mathcal{J}(\mathbb{R}^n)$ și $\lambda(\mathring{A}) = \lambda(\overline{A})$

3) $\operatorname{Fr} A \in \mathcal{J}(\mathbb{R}^n)$ și $\lambda(\operatorname{Fr} A) = 0$.

(P1) A nemărginită $\Rightarrow A \notin \mathcal{J}(\mathbb{R}^n)$

(P2) Fie $A \subseteq \mathbb{R}^n$ mărginită. Atunci:

$A \in \mathcal{J}(\mathbb{R}^2)$ cu $\lambda(A) = 0 \overset{def}{\Leftrightarrow} \lambda^*(A) = 0$

<u>DEM:</u> „$\Rightarrow$" $\lambda(A) = 0 \Rightarrow \lambda(A) = \lambda^*(A) = 0$

„$\Leftarrow$" Știm $0 \le \lambda_*(A) \le \underset{0}{\underset{\parallel}{\lambda^*(A)}}$

$\Rightarrow \lambda_*(A) = \lambda^*(A) = 0 \Rightarrow A \in \mathcal{J}(\mathbb{R}^2)$

$\Rightarrow \lambda(A) = 0$

<u>Corolar 1:</u>

Fie $A \subseteq \mathbb{R}^n$ mărginită.

A.î. $A \in \mathcal{J}(\mathbb{R}^n)$ cu $\lambda(A) = 0 \Leftrightarrow$

$\Leftrightarrow \forall \varepsilon > 0, \exists E \in \mathcal{E}(\mathbb{R}^n)$ o mulțime elementară cu $E \supset A$ și $vol(E) = \lambda(E) < \varepsilon$

<u>Corolar 2:</u>

Fie $A \subseteq B \subseteq \mathbb{R}^n$ 2 mulțimi cu $B \in \mathcal{J}(\mathbb{R}^n)$, $\lambda(B) = 0 \Rightarrow$

$\Rightarrow A \in \mathcal{J}(\mathbb{R}^n)$ și $\lambda(A) = 0$.

$\boxed{P3}$ Fie $f : [a,b] \to \mathbb{R}_+$ int(R).

Atunci

$\Gamma_f = \{(x,y) \in \mathbb{R}^2 \mid a \le x \le b, 0 \le y \le f(x)\} \in \mathcal{J}(\mathbb{R}^2)$ și $\lambda(\Gamma_f) = \int_a^b f(x)\,dx$

<u>SOL:</u>

Fie $\Delta = (a = x_0 < x_1 < \dots < x_m = b) \in \mathcal{D}([a,b])$

Fie $m_i = \inf\limits_{x \in [x_{i-1}, x_i]} f(x)$

$M_i = \sup\limits_{x \in [x_{i-1}, x_i]} f(x)$

-//-

Fie $E_1 = \bigcup_{i=1}^{n} [x_{i-1}, x_i] \times [0, m_i] \in \mathcal{E}(\mathbb{R}^2)$,

$E_1 \subset \Pi_f$

Fie $F_1 = \bigcup_{i=1}^{n} [x_{i-1}, x_i] \times [0, M_i] \in \mathcal{E}(\mathbb{R}^2)$

$F_1 \supset \Pi_f$

$$\lambda_*(A) = vol(E_1) = \sum_{i=1}^{n} (x_i - x_{i-1})(m_i - 0)$$

$$= \sum_{i=1}^{n} (x_i - x_{i-1}) \cdot m_i = s_f(\Delta)$$

$$\lambda^*(A) = S_f(\Delta)$$

<u>Deci</u>:

$$\lambda_*(A) = \sup_{\substack{E \subset A \\ E \in \mathcal{E}}} v(E) \geq vol(E_1) = s_f(\Delta)$$

$$\lambda^*(A) = \inf_{\substack{F \supset A \\ F \in \mathcal{E}}} v(F) \leq vol(F_1) = S_f(\Delta)$$

$$s_f(\Delta) \leq \lambda_*(A) \leq \lambda^*(A) \leq S_f(\Delta) \quad (1)$$

Dar din ipoteză f int (R):

$$\sup s_f(\Delta) = \inf S_f(\Delta) = \int_a^b f(x)\,dx$$

În $(1) \Rightarrow \lambda_*(A) = \lambda^*(A) = \int_a^b f(x)\,dx \Rightarrow$

$$\Rightarrow A \in \mathcal{J}(\mathbb{R}^2) \text{ și } \underset{\Pi_f}{\lambda(\underset{\shortparallel}{A})} = \int_a^b f(x)\,dx$$

<u>Consecința</u>: Fie $f, g : [a,b] \to \mathbb{R}$ cu f, g int (R) și $f(x) \leq g(x), \forall x \in [a,b]$. Atunci $\Pi_{f,g} = \{(x,y) \in \mathbb{R}^2 \mid a \leq x \leq b,\ f(x) \leq y \leq g(x)\} \in \mathcal{J}(\mathbb{R}^2)$ și

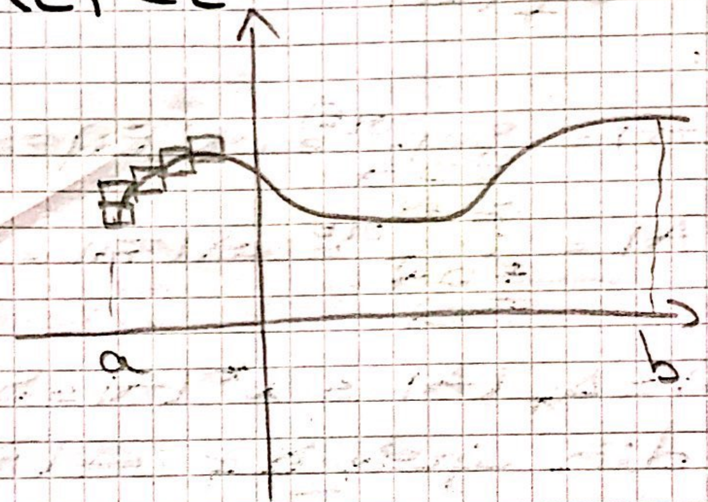
$$\lambda(\Pi_{f,g}) = \int_a^b (f(x) - g(x))\,dx.$$

$\boxed{P4}$ Fie $f:[a,b]\to\mathbb{R}$ int (R). Atunci $G_f=\{(x,y)\in\mathbb{R}^2 \mid x\in[a,b], y=f(x)\}\in\mathcal{J}(\mathbb{R}^2)$ și $\lambda(G_f)=0$

DEM:

Fie $\varepsilon>0$ fixat.

Vrem să aplicăm $\boxed{P2}$ adică să găsim $E\in\mathcal{E}(\mathbb{R}^2)$, $E\supset G_f$ și $vol(E)<\varepsilon$



Fie $\Delta=(a=x_0<x_1<\ldots<x_n=b)\in\mathcal{D}([a,b])$

Fie $m_i=\inf\limits_{x\in[x_{i-1},x_i]} f(x)$

$M_i=\sup\limits_{x\in[x_{i-1},x_i]} f(x)$

Fie $E:=\bigcup\limits_{i=1}^{n}[x_{i-1},x_i]\times[m_i,M_i]\in\mathcal{E}(\mathbb{R}^2)$ și $E\supset G_f$

Dar

$\lambda(E)=vol(E)=\sum\limits_{i=1}^{n}(x_i-x_{i-1})(M_i-m_i)$
$=S_f(\Delta)-s_f(\Delta)$

Dar din ip f int (R) $\overset{\text{Crit Darboux}}{\Longrightarrow}$ $\exists\,\delta_\varepsilon>0$ a.î. $S_f(\Delta)-s_f(\Delta)<\varepsilon$, $\forall\Delta\in\mathcal{D}([a,b])$ cu $\|\Delta\|<\delta_\varepsilon$ $\overset{\boxed{P2\ rel.1}}{\Longrightarrow}$ $G_f\in\mathcal{J}(\mathbb{R}^2)$ și $\lambda(G_f)=0$.

(P5) Dacă $A \in J(\mathbb{R}^p)$, $B \in J(\mathbb{R}^q) \Rightarrow$
$\Rightarrow A \times B \in J(\mathbb{R}^{p+q})$ și $\lambda(A \times B) =$
$= \lambda(A) \cdot \lambda(B)$.

Obs: Dc avem $B \notin J(\mathbb{R}^{\dots}) \Rightarrow$ nu putem ști nimic legat de natura mulțimii $A \times B$.

Ex) Studiați măsurabilitatea Jordan a mulțimilor următoare și aflați măsura celor măsurabile:

1) $A = \{(x,y) \in \mathbb{R}^2 \mid x + 2y = 0\}$
2) $B = [0,1] \cap \mathbb{Q}$
3) $C = ([0,1] \times [0,1]) \cap \mathbb{Q}^2$
4) $D_1 = [0,1] \times \{1\}$
   $D_2 = ([0,1] \setminus \mathbb{Q}) \times \{1\}$
5) $E_1 = \{a\} \subseteq \mathbb{R}$
   $E_2 = \{a_1, a_2, \dots, a_p\}$
   $E_3 = \{(a,b)\} \subseteq \mathbb{R}^2$
   $E_4 = \{\frac{1}{n} \mid n \in \mathbb{N}\} \subseteq \mathbb{R}$
6) Fie $f : [0,1] \to \mathbb{R}$, $f(x) = \begin{cases} 1, & x \in [0,1] \cap \mathbb{Q} \\ 0, & x \in [0,1] \setminus \mathbb{Q} \end{cases}$
   G_f, Π_f
7) $A = \{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 1, 0 \le y \le 1 - x^2\}$
8) $B = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 4\}$
9) $C = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1, y \ge x\}$
10) $D = \{(x,y) \in \mathbb{R}^2 \mid |x| \le 1, |y| \le x^2\}$
11) $f : [0,2] \to \mathbb{R}$, $f(x) = \begin{cases} \sin x, & x \in \mathbb{Q}, x \le 1 \\ x, & x \in \mathbb{Q}, x > 1 \\ -\cos x, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$
   G_f, Π_f

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

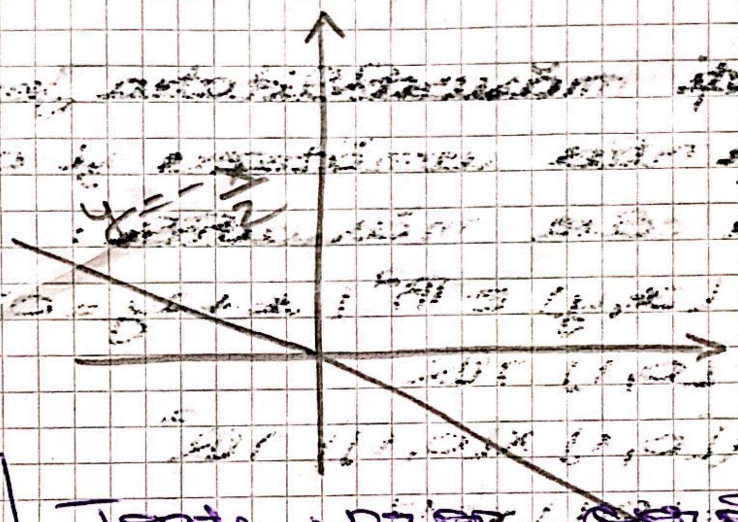
primul

al doilea

Sol:

1) $A = \{(x,y) \in \mathbb{R}^2 \mid x+2y=0\}$

$y = -\frac{x}{2}$.



!!! Teste grilă, ~~[illegible]~~ enun-
țate complete se vor enunța
în examen/lucrare.

$\boxed{P_1}$

A este nemărginită $\Rightarrow A \notin \mathcal{J}(\mathbb{R}^2)$

2) $B = [0,1] \cap \mathbb{Q}$

Folosim $\boxed{1}$:

$\underbrace{\mathring{B} = \emptyset}$: Pp prin RA că $\mathring{B} \neq \emptyset$

$\Rightarrow \exists\, x \in B,\ \exists\, r>0$ a.î. $B(x,r) = (x-r, x+r)$
$\subset B$ ~~$\not\subset$~~ ∃ cel puțin un nr $\notin \mathbb{Q}$

$\Rightarrow \lambda(\mathring{B}) = \lambda(\emptyset) = 0$. $\quad \emptyset \in \mathcal{E}(\mathbb{R})$

$B' = [0,1]$. Evident $\forall x \in [0,1], \forall V \in \mathcal{V}_x$,

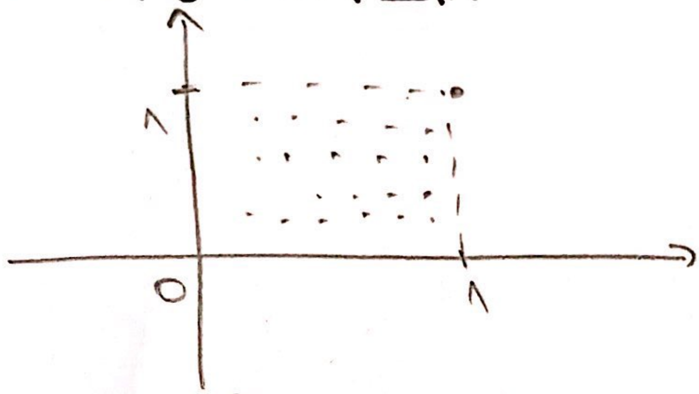
$V \cap B = \infty$ de elemente $\Rightarrow$

$\Rightarrow \overline{B} = B' \cup B = [0,1] \in \mathcal{E}(\mathbb{R}) \Rightarrow$

$\Rightarrow \lambda(\overline{B}) = \lambda([0,1]) = vol([0,1]) = 1$.

Deci $\lambda(\mathring{B}) \neq \lambda(\overline{B}) \Rightarrow B \notin \mathcal{J}(\mathbb{R})$.

3) $C = ([0,1]\times[0,1]) \cap \mathbb{Q}^2 = \{(x,y) \in \mathbb{R}^2 \mid x,y \in [0,1]\cap\mathbb{Q}\} \subseteq \mathbb{R}^2$



$\mathring{C} = \emptyset$: PP $\mathring{C} \neq \emptyset \Rightarrow \exists (x_0,y_0) \in C, \exists r>0$ a.î.

$B((x_0,y_0), r) \subset C$ ⨳

conține și pct de forma (x,y) cu $x,y \in \mathbb{R}\setminus\mathbb{Q}$ ⨳

($\mathbb{R}\setminus\mathbb{Q}$ densă în $\mathbb{R}^2$).

$\Rightarrow \lambda(\mathring{C}) = \lambda(\emptyset) = 0$ ($\emptyset \in \mathcal{J}(\mathbb{R}^2)$)

- $C' = [0,1]\times[0,1]$: $\forall (x,y)\in[0,1]\times[0,1]$, $\forall V \in \mathcal{V}_{(x,y)}$, $V\cap C = \infty$ de elem ($\mathbb{Q}^2$ densă în $\mathbb{R}^2$)

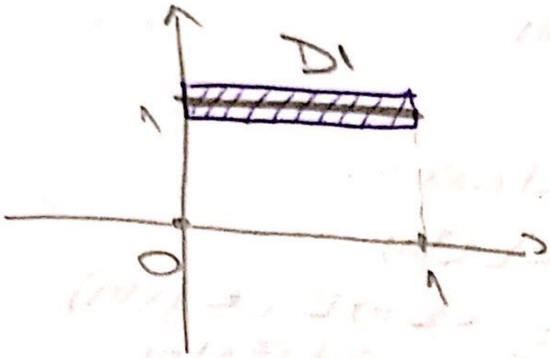
$\Rightarrow \overline{C} = C' \cup C = [0,1]\times[0,1]$

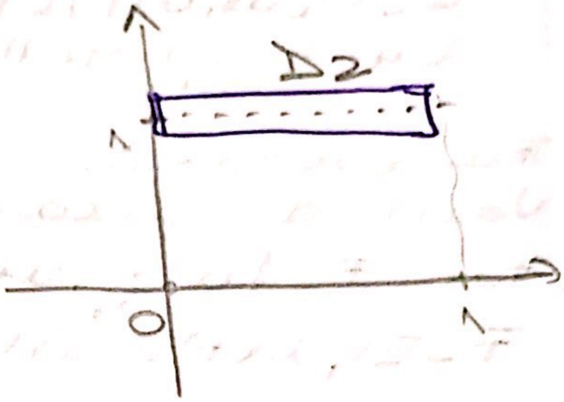
$\Rightarrow \lambda(\overline{C}) = \lambda([0,1]\times[0,1]) \overset{[0,1]\times[0,1]\in\mathcal{J}(\mathbb{R}^2)}{=} vol([0,1]\times[0,1]) = 1$.

$\Rightarrow \lambda(\mathring{C}) \neq \lambda(\overline{C}) \Rightarrow C \notin \mathcal{J}(\mathbb{R}^2)$.

4) $D_1 = [0,1]\times\{1\} = \{(x,1)\in\mathbb{R}^2 \mid x\in[0,1]\} \subseteq \mathbb{R}^2$

$D_2 = ([0,1]\setminus\mathbb{Q})\times\{1\}$





Sol 1: P2 Corolar 1.

Fie $\varepsilon>0$ fixat.

$E=[0,1]\times\left[1-\frac{\varepsilon}{4},1+\frac{\varepsilon}{4}\right]\in\mathcal{E}(\mathbb{R}^2)$

$D_1,D_2\subset E$ P2 cor 1

$\lambda(E)=vol(E)=1\cdot\frac{\varepsilon}{2}=\frac{\varepsilon}{2}<\varepsilon\Rightarrow$

$\Rightarrow D_1,D_2\in\mathcal{J}(\mathbb{R}^2)$ cu $\lambda(D_1)=\lambda(D_2)=0$.

Sol 2: $\boxed{P5}$

$D_1=\underset{\mathbb{R}}{\underset{\cap|}{[0,1]}}\times\underset{\mathbb{R}}{\underset{\cap|}{\{1\}}}$

$[0,1]\in\mathcal{E}(\mathbb{R}),\ \lambda([0,1])=vol([0,1])=1.$

$\{1\}\subseteq\mathbb{R}$; $E=[1-\varepsilon,1+\varepsilon]\in\mathcal{E}(\mathbb{R}),\ E\supset\{1\}$,

$vol(E)=\frac{\varepsilon}{2}<\varepsilon.\ \xRightarrow{\text{P2 corolar}}\ \{1\}\in\mathcal{J}(\mathbb{R}),$
$\lambda(\{1\})=0$

$\xRightarrow{P5}\ \lambda(D_1)=\lambda([0,1])\times\lambda(\{1\})=1\cdot0=0.$

Pentru $\boxed{D_2}$: nu mai merge P5!!!

Sol 3 Fie $f:[0,1]\to\mathbb{R}$, $f(x)=1$ const $\Rightarrow$

$\Rightarrow f$ int $(R)\ \xRightarrow{P4}\ D_1=Gr\,f\in\mathcal{J}(\mathbb{R}^2)$ şi $\lambda(D_1)=0$

Pentru $\boxed{D_2}$: nu merge P4!!!

Dar $D_2\subseteq D_1\in\mathcal{J}(\mathbb{R}^2)$, $\lambda(D_1)=0\Rightarrow$

$\xRightarrow{\text{P2 cor. 2}}\ D_2\in\mathcal{J}(\mathbb{R}^2),\ \lambda(D_2)=0.$

5) $E_1 = \{a\} \subseteq \mathbb{R}$

$E_2 = \{a_1, a_2, \dots, a_p\} \subseteq \mathbb{R}$

$E_3 = \{(a,b)\} \subseteq \mathbb{R}^2$

$E_4 = \{\frac{1}{n} \mid n \in \mathbb{N}\} \subseteq \mathbb{R}$

Fie $\varepsilon > 0$ fixat.

Vrem să aplicăm P2/cor 1:

E_1: Fie $F = [a - \frac{\varepsilon}{4}, a + \frac{\varepsilon}{4}] \in \mathcal{E}(\mathbb{R})$

$F \supset E_1$, $\lambda(F) = vol(F) = \frac{\varepsilon}{2} < \varepsilon \Rightarrow E_1 \in \mathcal{J}(\mathbb{R})$

$\lambda(E_1) = 0$

E_2: $F = [a_1 - \frac{\varepsilon}{4p}, a_1 + \frac{\varepsilon}{4p}] \cup [a_2 - \frac{\varepsilon}{4p}, a_2 + \frac{\varepsilon}{4p}] \cup \dots$

$\dots \cup [a_p - \frac{\varepsilon}{4p}, a_p + \frac{\varepsilon}{4p}] \in \mathcal{J}(\mathbb{R})$

$E_2 \subset F$; $\lambda(F) = vol(F) = 2\varepsilon p / 4p = \frac{\varepsilon}{2} < \varepsilon$

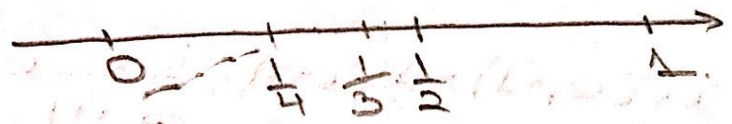
$\xRightarrow{P2/cor 1} E_2 \in \mathcal{J}(\mathbb{R})$ și $\lambda(E_2) = 0$

E_3: $F = [a - \frac{\sqrt{\varepsilon}}{4}, a + \frac{\sqrt{\varepsilon}}{4}] \times [b - \frac{\sqrt{\varepsilon}}{4}, b + \frac{\sqrt{\varepsilon}}{4}] \in \mathcal{E}(\mathbb{R}^2)$

$F \supset E_3$, $\lambda(F_0) = vol(F) = \left(2\frac{\sqrt{\varepsilon}}{4}\right)^2 = \frac{\varepsilon}{4} < \varepsilon \Rightarrow$

$\xRightarrow{P2/cor 1} E_3 \in \mathcal{J}(\mathbb{R}^2)$ cu $\lambda(E_3) = 0$.

E_4



$\left.\begin{array}{l} I_0 = [-\varepsilon, \varepsilon] \\ \frac{1}{n} \searrow 0 \end{array}\right| \Rightarrow \exists N \in \mathbb{N}$ a.î. $\frac{1}{n} \in I_0$, $\forall n \in \mathbb{N}$.

Fie $I_1 = [1-\varepsilon, 1+\varepsilon]$, $I_2 = [\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon]$,

$\dots, I_{N-1} = [\frac{1}{N-1} - \varepsilon, \frac{1}{N-1} + \varepsilon]$.

$\Rightarrow F = I_0 \cup I_1 \cup \dots \cup I_{N-1} \in \mathcal{E}(\mathbb{R})$

$F \supset E_4$

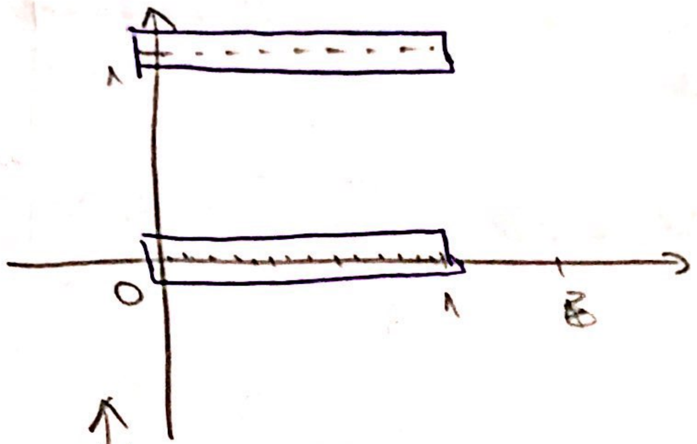
$\lambda(F) = vol(F) = \underbrace{2\varepsilon + 2\varepsilon + \dots + 2\varepsilon}_{N} = 2\varepsilon N \Rightarrow$

$\Rightarrow E_4 \in \mathcal{J}(\mathbb{R})$ cu $\lambda(E_4) = 0$.

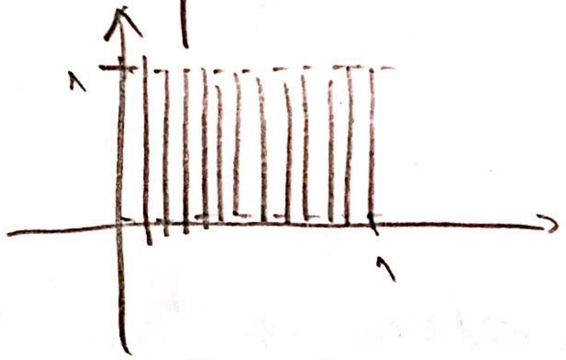
6) Fie $f:[0,1]\to\mathbb{R}$, $f(x)=\begin{cases}1, & x\in[0,1]\cap\mathbb{Q}\\ 0, & x\in[0,1]\setminus\mathbb{Q}\end{cases}$

G_f, π_f

G_f:



π_f:



Fie $\varepsilon>0$ fixat.

$E=([0,1]\times[-\varepsilon,\varepsilon])\times([0,1]\times[1-\varepsilon,1+\varepsilon])\in\mathcal{E}(\mathbb{R}^2)$

$E\supset G_f$, $\lambda(E)=vol(E)=1\cdot 2\varepsilon+1\cdot 2\varepsilon=4\varepsilon\Rightarrow$

P $\varepsilon>0$ $\Rightarrow G_f\in\mathcal{J}(\mathbb{R}^2)$, $\lambda(G_f)=0$.

$\underline{\mathring{\pi}_f=\emptyset}$: pp $\mathring{\pi}_f\neq\emptyset\Rightarrow\exists(x_0,y_0)\in\mathring{\pi}_f, \exists r>0$ a.î. $B((x_0,y_0),r)\subset\pi_f$ ⨯.

$\not\exists$ cel puțin un element de forma (x,y) cu $x,y\in$ ~~$\mathbb{Q}\cap[0,1]$~~ $[0,1]\setminus\mathbb{Q}$ ($\mathbb{Q}$ densă în $\mathbb{R}$).

$\underline{\pi_f'=[0,1]\times[0,1]}$: Într-adevăr $\forall(x,y)\in[0,1]\times[0,1]$, $\forall V\in\mathcal{V}(x,y)$, $V\cap\pi_f=\infty$ de elemente ($\mathbb{Q}^2$ densă în $\mathbb{R}^2$)

$\Rightarrow\overline{\pi_f}=\pi_f'\cup\pi_f=[0,1]\times[0,1]\Rightarrow$

$\Rightarrow\lambda(\overline{\pi_f})=\lambda([0,1]\times[0,1])\overset{[0,1]\in\mathcal{E}(\mathbb{R}^2)}{=}vol([0,1]\times[0,1])=1$

$\Rightarrow\lambda(\mathring{\pi}_f)\neq\lambda(\overline{\pi_f})\Rightarrow\pi_f\notin\mathcal{J}(\mathbb{R}^2)$.